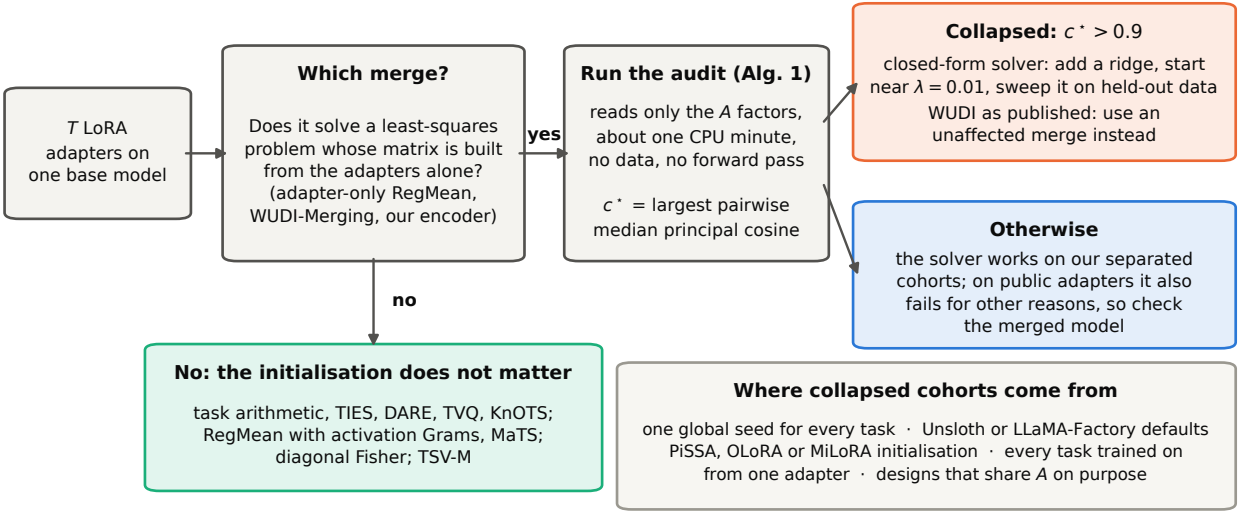
T LoRA adapters on one base model
Which merge?
Does it solve a least-squares problem whose matrix is built from the adapters alone? (adapter-only RegMean, WUDI-Merging, our encoder)
yes
no
Run the audit (Alg. 1)
reads only the A factors, about one CPU minute, no data, no forward pass
$c^\star$ = largest pairwise median principal cosine
Collapsed: $c^\star > 0.9$
closed-form solver: add a ridge, start near $\lambda = 0.01$, sweep it on held-out data
WUDI as published: use an unaffected merge instead
Otherwise
the solver works on our separated cohorts; on public adapters it also fails for other reasons, so check the merged model
No: the initialisation does not matter
task arithmetic, TIES, DARE, TVQ, KnOTS; RegMean with activation Grams, MaTS; diagonal Fisher; TSV-M
Where collapsed cohorts come from
one global seed for every task · Unsloth or LLaMA-Factory defaults
PiSSA, OLoRA or MiLoRA initialisation · every task trained on from one adapter · designs that share A on purpose